

SECURITY CODE Lock

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The circuit described here is of an electronic combination lock. It opens only when you press four keys in the correct sequence (referred to as a code) out of a total of eight keys installed at your entrance door. If you press the wrong key sequence, the circuit resets and the lock will not open. The author's prototype is shown in Fig. 1.

Circuit and working

The circuit diagram of the security code lock is shown in Fig. 2. The circuit consists of step-down transformer X1, bridge rectifier BR1, 12V voltage regulator LM7812 (IC1), dual-D flip-flop ICs 4013 (IC2 and IC3), rectifier diode 1N4007 (D1), pnp transistor BC557 (T1), npn transistor 2N2219 (T2), 12V single changeover relay RL1, and a few other components.

This project operates with 12V DC which is derived from the 230V AC primary to 15V, 500mA secondary step-down transformer X1. The 230V AC mains is connected to the primary of X1 via CON1 in the circuit. The secondary of X1 is 15V AC, which is connected to bridge rectifier BR1 for rectification and filtered by capacitor C1. The rectified and filtered voltage is then supplied to regulator IC

LM7812 (IC1) to regulate it to 12V for enabling the circuit. The glowing LED1 indicates that 12V is available to enable the circuit.

The heart of this code lock circuit is the CD4013 dual-D flip-flop IC. The clock pins (pins 3 and 11 of IC2 and IC3) of its four flip-flops are connected to CO1, CO2, CO3, and CO4 points. The correct code sequence for energising relay RL1 is achieved by clocking in the correct sequence, which is the code of the lock. In this case, the code is 8, 2, 6, and 5 (8265). Therefore, you need to press switches S8, S2, S6, and S5 in the correct sequence to open the door. The remaining four switches (S1, S3, S4, and S7) are connected to CO5 reset point, which resets all the flip-flops. By momentarily pressing switches S8, S2, S6, and S5, the clock input pin goes high and alters the state of the flip-flops. The Q output pin of each flip-flop is wired to the D input pin of the next flip-flop, while the D pin (pin 5) of the first flip-flop (IC2) is grounded.

The lock code can be changed by altering the connections to the points CO1, CO2, CO3, and CO4. For example, if you want to set the code as 1756, you should connect

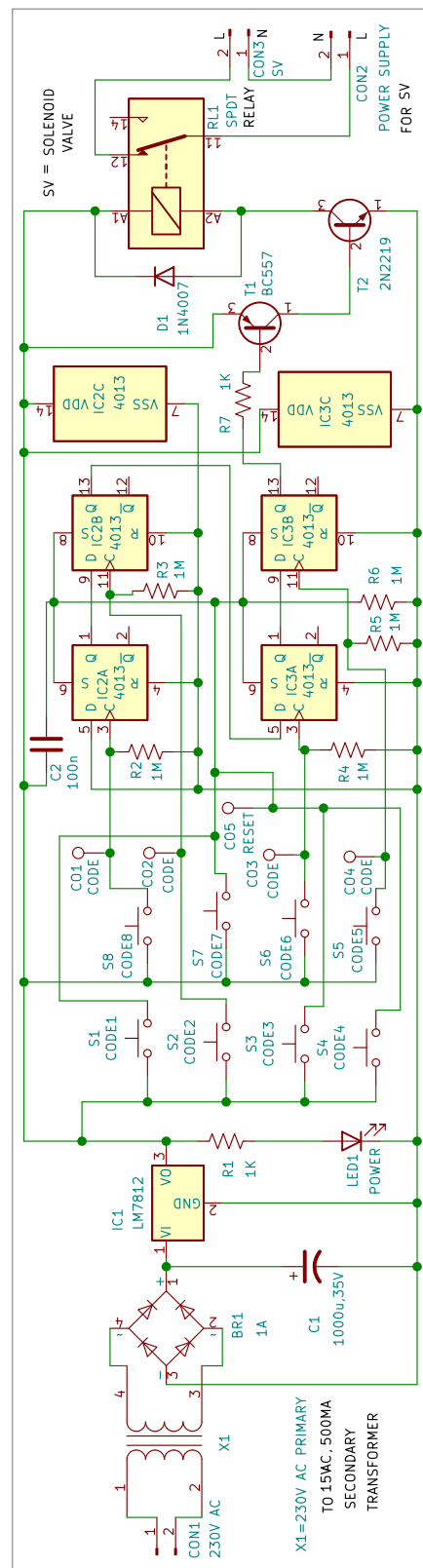


Fig. 2: Circuit diagram

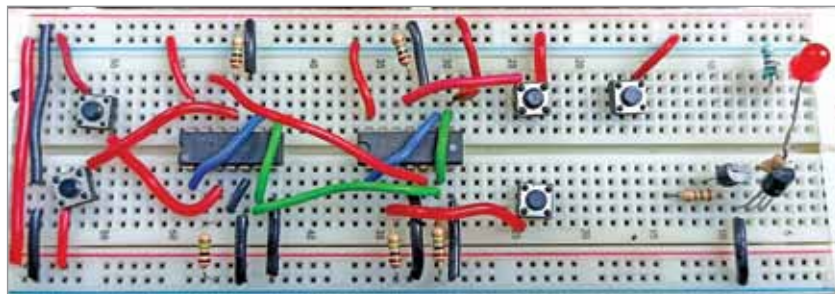


Fig. 1: Prototype of the lock tested on a breadboard